

WE Prerequisite: Indoor Water Use Reduction

This prerequisite applies to

- ▶ BD+C: New Construction
- ▶ BD+C: Core & Shell
- ▶ BD+C: Schools
- ▶ BD+C: Retail
- ▶ BD+C: Data Centers
- ▶ BD+C: Warehouses & Distribution Centers
- ▶ BD+C: Hospitality
- ▶ BD+C: Healthcare

INTENT

To reduce indoor potable water consumption and preserve no and low cost potable water resources.

REQUIREMENTS

NC, CS, SCHOOLS, NC-RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, NC-HOSPITALITY, HEALTHCARE

Building Water Use

For the fixtures and fittings listed in Table 1, as applicable to the project scope, reduce aggregate water consumption by 20% from the baseline. Base calculations on the volumes and flow rates shown in Table 1.

All newly installed toilets, urinals, private lavatory faucets, and showerheads that are eligible for labeling must be WaterSense labeled (or a local equivalent for projects outside the U.S.).

Table 1. Baseline water consumption of fixtures and fittings

<i>Fixture or fitting</i>	<i>Baseline (IP units)</i>	<i>Baseline (SI units)</i>
Toilet (water closet)*	1.6 gpf	6 lpf
Urinal*	1.0 gpf	3.8 lpf
Public lavatory (restroom) faucet	0.5 gpm at 60 psi** all others except private applications	1.9 lpm at 415 kPa, all others except private applications
Private lavatory faucets	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Kitchen faucet (excluding faucets used exclusively for filling operations)	2.2 gpm at 60 psi	8.3 lpm at 415 kPa
Showerhead*	2.5 gpm at 80 psi per shower stall	9.5 lpm at 550 kPa per shower stall

* WaterSense label available for this product type
gpf = gallons per flush
gpm = gallons per minute
psi = pounds per square inch

lpf = liters per flush
lpm = liters per minute
kPa = kilopascals

Projects located where standard supply pressure is different than the LEED baseline supply pressure may calculate the water consumption of flow fixtures and fittings at the local standard supply pressure.

Appliance and Process Water Use

Install appliances, equipment, and processes within the project scope that meet the requirements listed in the tables below.

Existing appliances intended for reuse in the project are not required to meet the requirements in Table 2.

Table 2. Standards for appliances

<i>Appliance</i>	<i>Requirement</i>
Residential clothes washers	ENERGY STAR or performance equivalent*
Commercial clothes washers	ENERGY STAR or performance equivalent
Residential dishwashers (standard and compact)	ENERGY STAR or performance equivalent*
Prerinse spray valves	≤ 1.3 gpm (4.9 lpm)
Ice machine	ENERGY STAR or performance equivalent and use either air-cooled or closed-loop cooling, such as chilled or condenser water system

gpm = gallons per minute

lpm = liters per minute

*Projects in Europe may install residential appliances meeting the EU A label.

Table 3. Standards for processes

<i>Process</i>	<i>Requirement</i>
Heat rejection and cooling	No once-through cooling with potable water for any equipment or appliances that reject heat
Cooling towers and evaporative condensers	Equip with • makeup water meters • conductivity controllers and overflow alarms • efficient drift eliminators that reduce drift to maximum of 0.002% of recirculated water volume for counterflow towers and 0.005% of recirculated water flow for cross-flow towers

Healthcare, Retail, Schools, and Hospitality Only

In addition, water-consuming appliances, equipment, and processes must meet the requirements listed in Tables 4 and 5.

Table 4. Standards for appliances

<i>Kitchen equipment</i>	<i>Requirement (IP units)</i>	<i>Requirement (SI units)</i>
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Dishwasher	Undercounter	≤ 1.6 gal/rack	≤ 6.0 liters/rack
	Stationary, single tank, door	≤ 1.4 gal/rack	≤ 5.3 liters/rack
	Single tank, conveyor	≤ 1.0 gal/rack	≤ 3.8 liters/rack
	Multiple tank, conveyor	≤ 0.9 gal/rack	≤ 3.4 liters/rack
	Flight machine	≤ 180 gal/hour	≤ 680 liters/hour
Food steamer	Batch	≤ 6 gal/hour/pan	≤ 23 liters/hour/pan
	Cook-to-order	≤ 10 gal/hour/pan	≤ 38 liters/hour/pan
Combination oven,	Countertop or stand	≤ 3.5 gal/hour/pan	≤ 13 liters/hour/pan
	Roll-in	≤ 3.5 gal/hour/pan	≤ 13 liters/hour/pan

Table 5. Process requirements

Discharge water temperature tempering	Where local requirements limit discharge temperature of fluids into drainage system, use tempering device that runs water only when equipment discharges hot water OR Provide thermal recovery heat exchanger that cools drained discharge water below code-required maximum discharge temperatures while simultaneously preheating inlet makeup water OR If fluid is steam condensate, return it to boiler
Venturi-type flow-through vacuum generators or aspirators	Use no device that generates vacuum by means of water flow through device into drain

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

WE prerequisite Indoor Water Use Reduction incorporates feedback from and market barriers identified by LEED project teams. The intent of these changes is to make prerequisite compliance more achievable and relevant to international project teams by recognizing regional variations while maintaining the stringency of requirements for water performance.

Further Explanation

Refer to the LEED v4 reference guide, with the following additions:

Occupant Types

If the project includes separate gender neutral and/or ADA restrooms without urinals, assume that 5% of male occupants and 5% of female occupants use these restrooms. Enter 95% into the percent of males expected to use restrooms with urinals in the Indoor Water Use Reduction Calculator. Alternately, estimate this percentage based on the project's restroom layout and anticipated usage patterns or weighted fixture counts.

International Tips

Refer to the LEED v4 reference guide, with the following modifications:

- ▶ International projects located where standard water supply pressure is different than the LEED baseline supply pressure values may calculate the water consumption of flow fixtures and fittings at the local standard water supply pressure. Product cutsheets must demonstrate that the fixture or fitting complies with the LEED baseline flow rate.
 - Projects that are unable to provide manufacturer documentation of the fixture flow rate at the LEED baseline water supply pressure may use the local standard supply water pressure in the design case and the standard LEED baseline flow rates in the baseline case.
 - Projects served by water supply pressures different than specified in LEED may install pressure compensating aerators in flow fixtures to achieve the desired flow rate without compromising user satisfaction.
- ▶ Where local building code conflicts with LEED prescriptive requirements, project teams may select fixtures that comply with local code and compensate by selecting more water-efficient fixtures for other fixture types, provided that the flow rate for the fixture is the lowest allowable by code, and the project meets in aggregate the required 20% reduction from LEED baseline. Provide an excerpt of the local code highlighting the flow rate requirements when submitting for review.
- ▶ The EU A label for residential appliances is an acceptable alternative to ENERGY STAR. The EU Ecodesign and Labelling framework establishes minimum performance standards for the energy and environmental performance of appliances and products.
- ▶ The CEE Commercial Clothes Washer Specification is no longer active. Commercial clothes washers require the ENERGY STAR label or performance equivalent.
 - Performance equivalent refers to both the energy and water criteria in the ENERGY STAR product specifications, available on the ENERGY STAR website.
 - Products must meet the standards of the current version of ENERGY STAR as of the date of their purchase.

Referenced Standards

Refer to the LEED v4 reference guide, with the following modification:

Delete "Consortium for Energy Efficiency" from the list of referenced standards.

Connection to Ongoing Performance

- ▶ LEED O+M WE credit Water Performance: Selecting efficient fixtures, fittings and appliances in the design phase helps ensure that projects will use less water throughout the building life cycle, which may help improve a project's water performance score.